

MULTILINGUAL EXPRESSIVE TEXT-TO-SPEECH (TTS) SYSTEM

Project Report

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Abstract

The Multilingual Expressive Text-to-Speech (TTS) System converts text into natural-sounding speech in **English, Hindi, Telugu, and Kannada** using React, FastAPI, Piper TTS, and open-source local TTS models. The system supports fully offline speech generation after the required models and dependencies are installed. It also supports PDF and TXT for speech conversion.

Objectives

Develop a multilingual TTS system, support four languages, convert text/PDF/TXT to speech, provide a responsive interface, and allow audio playback and download.

Technologies Used

Frontend: React.js

Backend: FastAPI

Language: Python

TTS: Piper TTS and MMS VITS local models

Libraries: Axios, PyPDF2, python-multipart, Transformers, PyTorch

System Architecture

User → React UI → FastAPI → Local TTS Models → WAV Audio → Play/Download

Features

Text-to-Speech, PDF-to-Speech, TXT-to-Speech, English, Hindi, Telugu, Kannada, Character Counter, Word Counter, Copy, Clear, Loading Indicator, Audio Player, Download, Responsive UI.

Implementation

React provides the web interface. FastAPI processes speech-generation requests and uses locally installed open-source TTS models for English, Hindi, Telugu, and Kannada. The generated speech is returned as a WAV audio file. PDF and TXT files are processed locally to extract text before speech generation.

Challenges

One of the main challenges was finding suitable open-source local models for regional Indian languages, especially Kannada. This was addressed by integrating a local MMS VITS Kannada model and testing the generated audio independently. The final system supports all four required languages without relying on external TTS APIs.

Results

The application successfully converts text, PDF, and TXT content into speech in English, Hindi, Telugu, and Kannada. The generated audio can be played and downloaded through the web interface. All core speech generation is performed locally.

Conclusion

The project meets the required objectives by providing a multilingual, expressive, and offline Text-to-Speech solution using open-source local models. The system supports English, Hindi, Telugu, and Kannada along with PDF/TXT input, audio playback, and download functionality.

Future Enhancements

Support additional Indian languages, improve expressive speech quality, and add advanced speed, pitch, and emotion controls.

References

- React Documentation
- FastAPI Documentation
- Piper TTS Documentation
- Microsoft Speech / MMS (Massively Multilingual Speech) Documentation
- PyTorch Documentation
- Hugging Face Transformers Documentation
- PyPDF2 Documentation

Appendix: web Application

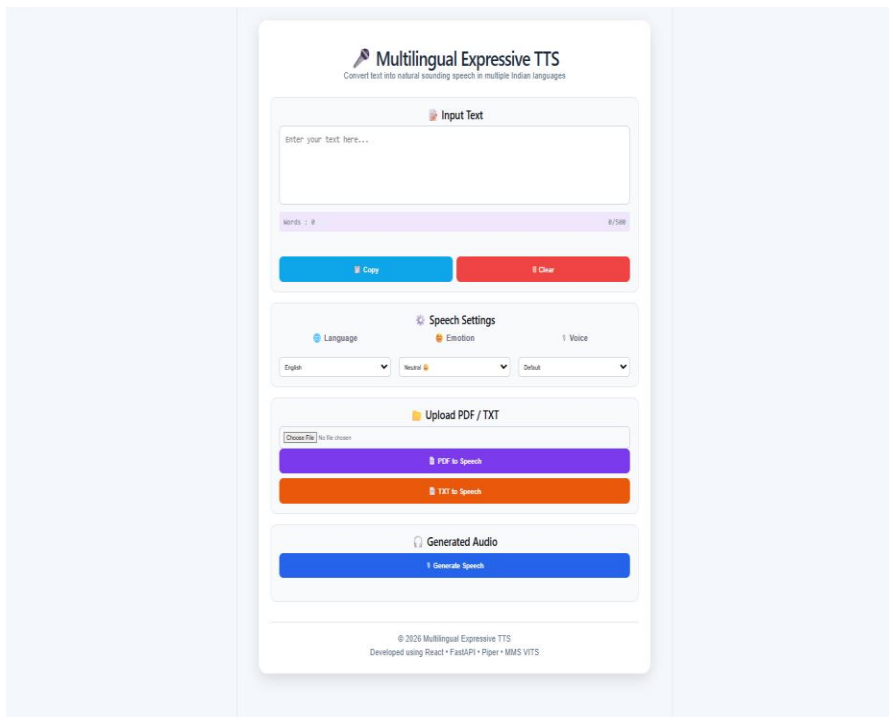


Figure 1: Web Interface of the Multilingual Expressive TTS System